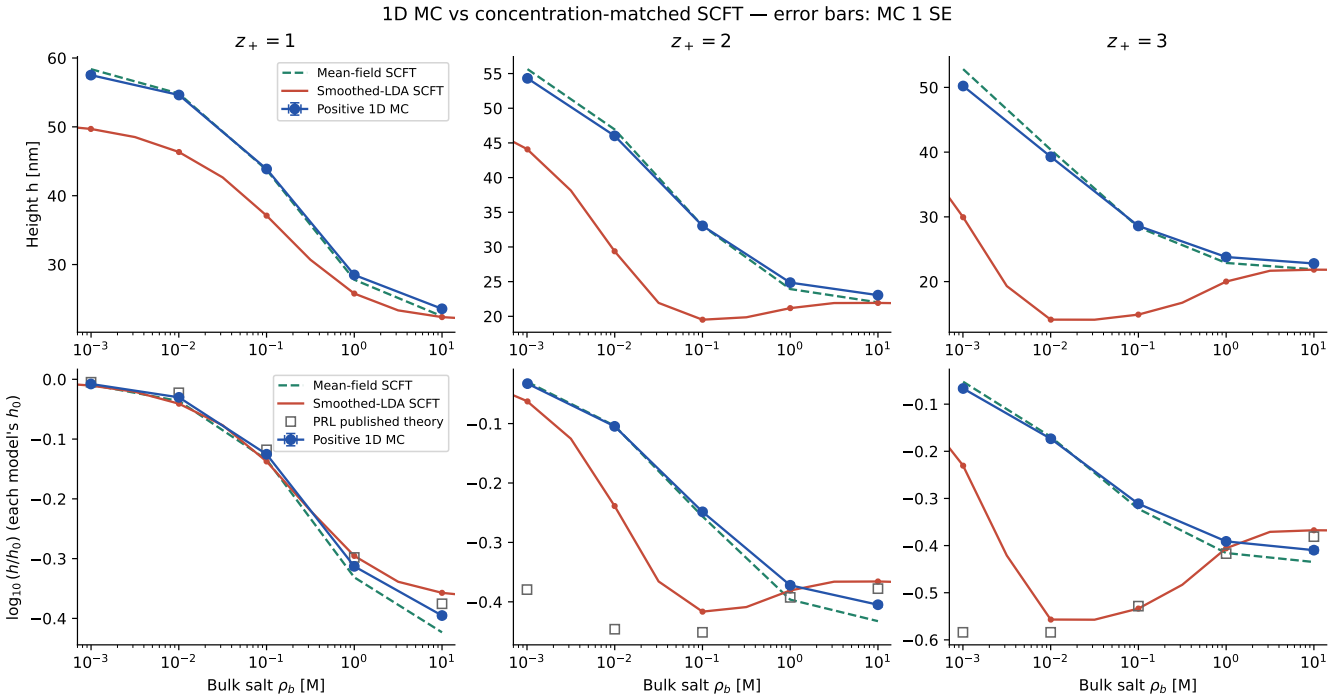
Sampling versus PRL-type SCFT

Matched physical parameters and measured bulk concentrations / 2026-09-07

Result: the 1D samples are closer to mean-field SCFT with correlations disabled. Blue denotes actual MC samples; red denotes the existing PRL-type smoothed-LDA SCFT; green denotes SCFT with correlation self-energies disabled, retaining the mean potential and canonical C. Black squares are digitized paper theory curves, distinct from our LDA implementation.



Top: absolute height $h = 2\int z\rho_P\,dz/(\sigma N)$. Bottom: $\log_{10}(h/h_0)$, using each model's independently computed salt-free height. MC error bars are 1 SE; the shared h_0 uncertainty correlates the normalized points. Red lines show the earlier 19-point scan; small red dots are new solutions at the MC concentrations.

Heights near 0.01 M [nm]

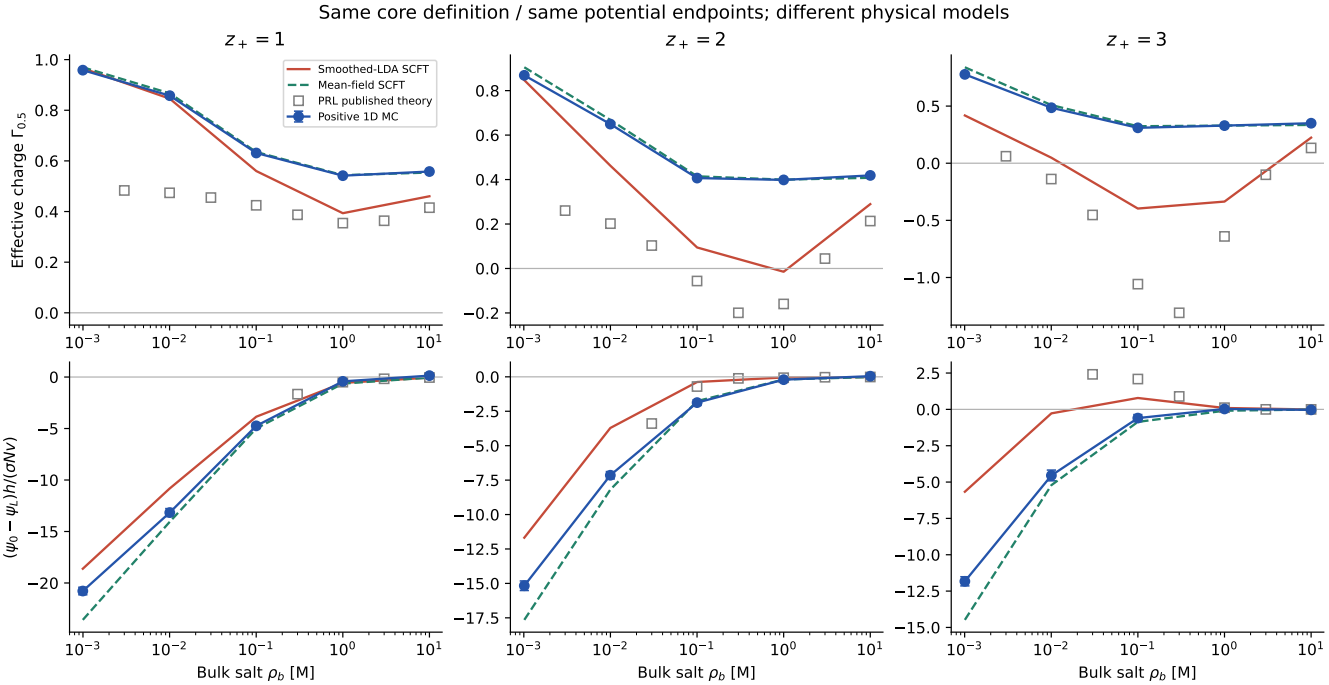
SCFT uses the actual MC bath concentration for each valence.

z_+	MC ± SE	LDA SCFT	Mean-field SCFT
1	54.63 ± 0.26	46.34	54.83
2	46.02 ± 0.21	29.38	46.94
3	39.28 ± 0.35	14.12	40.37

Salt-free heights are MC 58.561 nm, LDA 50.911 nm, and mean field 59.602 nm. These reference heights were not fitted to one another. MC and mean field do not coincide: the largest absolute MC–MF height difference across all comparison points is 2.607 nm; some differences exceed the MC statistical errors.

$N = 100$, $b = 1\,\text{nm}$, $v = 1\,\text{nm}^3$, $\sigma = 0.1\,\text{nm}^{-2}$, $\chi = 0.5$, $\ell_B = 0.7\,\text{nm}$, $a = 0.25\,\text{nm}$, and $L = 200\,\text{nm}$ were retained. Replacing the nominal concentration with the measured MC bath concentration changes LDA heights by at most 0.053 nm, insufficient to explain the large observed differences. The paper uses a Gibbs dividing surface, so the normalized plot also compares different height definitions.

Charge, potential, and model differences



$\overline{\Gamma}_{0.5}$ is the salt-excess ratio in the core above half the polymer peak density. SCFT densities were averaged over four cells onto the MC grid; potentials were compared at $z = 0.25$, 199.75 nm. This avoids mistaking an absolute gauge difference for potential reversal. Colors and symbols are as above.

Trivalent salt near 0.1 M: different charge and potential signs

Observable	MC ± SE	LDA SCFT	Mean-field SCFT
$\overline{\Gamma}_{0.5}$	0.310 ± 0.004	-0.397	0.324
Ψ_{rel}	-0.599 ± 0.244	0.784	-0.860

A 1D mean profile can retain 3D fluctuations

The PRL correlation theory includes a 3D charge Green function $G(\mathbf{r}, \mathbf{r}')$. It retains transverse modes even for a planar mean density $\rho(z)$. Its planar Fourier representation is

$$G(\mathbf{r}, \mathbf{r}') = \int \frac{d^2 q_\parallel}{(2\pi)^2} e^{i\mathbf{q}_\parallel \cdot (\mathbf{r}_\parallel - \mathbf{r}'_\parallel)} G_{q_\parallel}(z, z')$$

The current MC samples a planar Coulomb action with only $q_\parallel = 0$, excluding transverse fluctuations. Increasing the sample count cannot restore those excluded correlation modes. Our LDA approximates local 3D ionic correlations; equivalence to the paper's full nonlocal theory remains unverified.

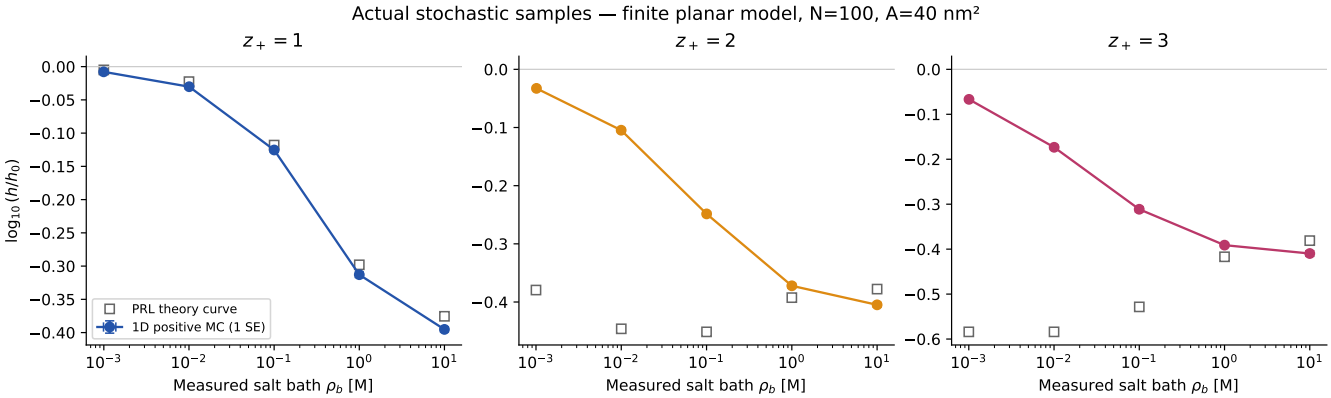
This is a **comparison of different models**, not a test of MC algorithms for one action. MC and MF also differ in finite A, integer cell solvent entropy, grid, charge smearing, and 1D fluctuations. The PRL neglects u_P^{ex} but retains u_P^0; that omission alone cannot explain the discrepancy. Theory: Duan et al., Eqs. (3)–(5) and SI Sec. II. All new SCFT states passed full residual 10^{-7} and independent chain density/contour Q checks at 10^{-10}. Numbers, density profiles, and run paths: `SAMPLING_SCFT_COMPARISON.md`. The SCFT runs used the existing research runner; production FFT/C++/CUDA and MC/CL code were unchanged.

N=100 brush: Monte Carlo sampling

2026-09-07 / New stochastic calculations, separate from the earlier Fig. 2–3 SCFT scan

Sample averages were obtained by stochastic updates of incompressible P/S and ion configurations and the real field in the attractive channel. Parameters were $N = 100$, $b = 1\,\mathrm{nm}$, $\sigma = 0.1\,\mathrm{nm}^{-2}$, $v = 1\,\mathrm{nm}^3$, $\chi = 0.5$, $\ell_B = 0.7\,\mathrm{nm}$, and $a = 0.25\,\mathrm{nm}$.

Scope: a finite planar model retaining only z, with 4 chains in $A = 40\,\mathrm{nm}^2$, $L = 200\,\mathrm{nm}$, and $M = 400$ cells. Transverse charge fluctuations are excluded, so this differs from the paper's 3D statistical model. No LDA self-energy or PB/SCFT solution was inserted into the samples.



Circles and error bars: sample means and 1 SE. Open squares: digitized PRL theory curves. The horizontal axis uses concentrations measured in separate pure-salt bulk simulations. The table lists nominal targets.

Target ρ_b [M]	$z_+ = 1$: h [nm]	$z_+ = 2$: h [nm]	$z_+ = 3$: h [nm]
0.001	57.52 ± 0.36	54.31 ± 0.22	50.21 ± 0.29
0.01	54.63 ± 0.26	46.02 ± 0.21	39.28 ± 0.35
0.1	43.88 ± 0.17	33.04 ± 0.19	28.60 ± 0.33
1	28.49 ± 0.21	24.86 ± 0.17	23.80 ± 0.19
10	23.57 ± 0.14	23.05 ± 0.17	22.80 ± 0.29

$$h = \frac{2\int z\langle\rho_P(z)\rangle\,dz}{\sigma N}.$$

The salt-free reference was also sampled directly: $h_0 = 58.56 \pm 0.35\,\mathrm{nm}$. Height is explicitly defined by the first moment; it is not automatically equivalent to the paper's Gibbs dividing-surface procedure. Height decreases at the sampled concentrations, without reproducing the paper's pronounced re-expansion.

Sampled distribution and validation

The cell capacity is $K = A\Delta z/v = 20$. Polymer paths retain labeled bead coordinates; solvent is eliminated exactly through $s_j = K - p_j$. Monovalent counterions are fixed at $\sum_j c_j = n_{\text{chain}}N = 400$, while salt satisfies $\sum_j n_{-,j} = z_+ \sum_j n_{+,j}$.

$$\pi \propto B(\boldsymbol{R}) \exp[-U_e - U_\chi] \prod_j \frac{a_+^{n_{+,j}} a_-^{n_{-,j}}}{s_j!\, c_j!\, n_{+,j}!\, n_{-,j}!},$$

$$U_\chi = \frac{\chi}{K} \sum_j p_j(K - p_j), \qquad U_e = \frac{2\pi \ell_B \Delta z}{A} \sum_e Q_e^2,$$

$$Q_e = \sum_j F_{ej}[-p_j + c_j + z_+ n_{+,j} - n_{-,j}].$$

B is the weight of Gaussian bonds and a graft source of width 0.5 nm. F_{ej} is the cumulative distribution of Gaussian charge of width $a/\sqrt{\pi}$. The Gaussian electrostatic field is integrated exactly, including finite-grid self terms. Walls use Neumann electric boundary conditions; periodic bulk calculations replace Q_e by $Q_e - \overline{Q}$.

Real fields are drawn from their conditional distribution

$$e^{(\chi/K)\sum_j (p_j - \bar{p})^2} \propto \int_{\sum_j x_j = 0} d\boldsymbol{x}\, e^{-K\sum_j x_j^2/(4\chi) + \sum_j p_j x_j},$$

$$\boldsymbol{x} \mid \boldsymbol{p} \sim \mathcal{N}\left(\frac{2\chi}{K}(\boldsymbol{p} - \bar{p}), \frac{2\chi}{K}\left[I - \frac{\mathbf{1}\mathbf{1}^T}{M}\right]\right).$$

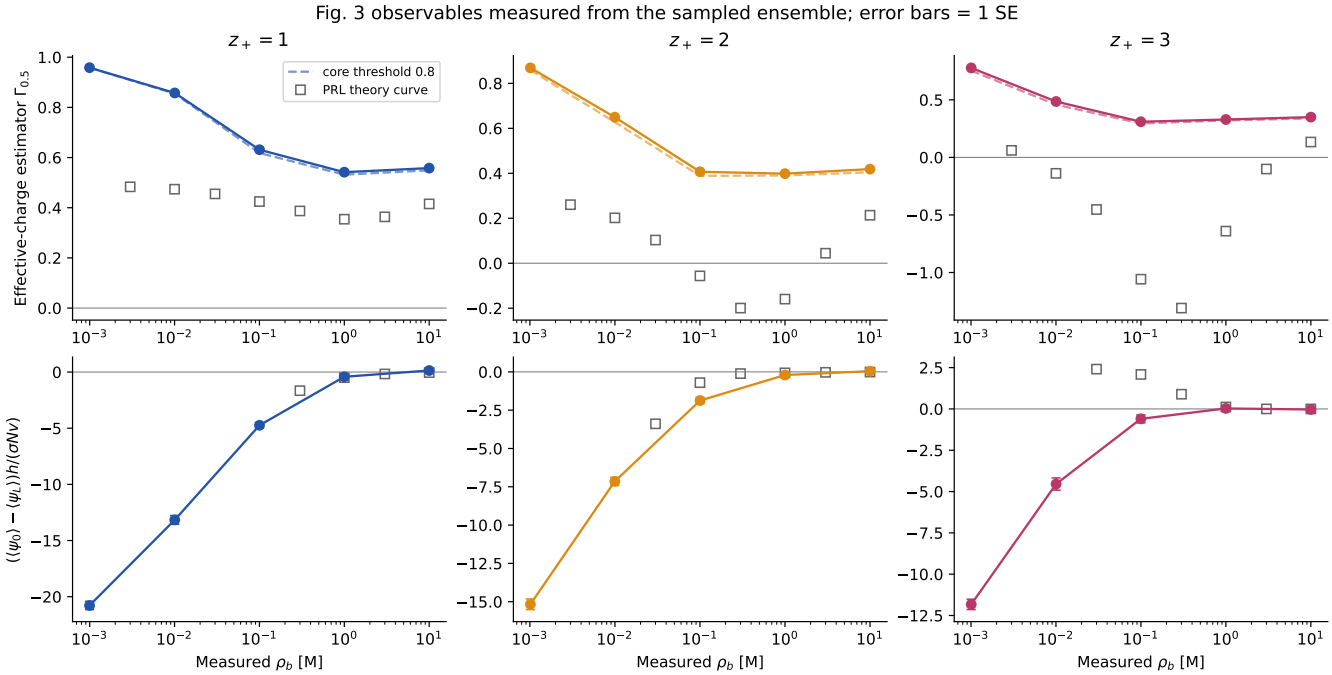
Particle positions use Metropolis updates; ions use binomial conditional redistribution proposals; salt counts use neutral-cluster insertion and deletion. Same-cell and separated-cell proposals are mixed, including each reverse proposal probability. Bulk chemical potentials are adjusted only during warmup and held fixed during measurement and brush sampling. The largest relative bulk concentration SE is 0.941%.

Validation and statistical scope

Polymer paths, canonical C, and neutral salt configurations were summed directly for 2 cells and 2 beads. The relative partition error bound from truncating salt counts is below 3×10^{-40}. Across 12 runs ($z_+ = 1, 2, 3$, two methods, two seeds), all checked means were within 2.67 SE of exact values. Detailed balance and energy updates were checked against independent full weights.

Each of 32 standard brush runs, 2 particle-only controls, and 8 grid/area controls used 200,000 warmup sweeps and 1,000,000 measurement sweeps. For divalent salt at 0.01 M, initial split-$\widehat{R} = 1.095$ and relaxation from the compact start motivated 1,000,000 extra sweeps for both seeds; the table uses these extensions. The original intervals are preserved. Each sweep includes one random single-bead proposal per bead on average. Errors account for autocorrelation and 10/20/40-block analysis. Per-run height ESS ranges from 10.1 to 79.4. The maximum split-$\widehat{R}$ across standard pairs is 1.039; the largest height difference from particle-only controls is 1.63 SE. **Update counts are not independent sample counts.**

Fig. 3 from sample averages



Solid: MC means with core threshold 0.5. Dotted: threshold 0.8. Open squares: PRL theory curves. Surface potential uses the mean difference between first/last cell centers, without rescaling to fit the paper.

$$\overline{\Gamma}_c = 1 - z_+ \frac{\int_{\langle\rho_P\rangle > c\max\langle\rho_P\rangle} [\langle\rho_+\rangle - \rho_b]\, dz}{\int_{\langle\rho_P\rangle > c\max\langle\rho_P\rangle} \langle\rho_P\rangle\, dz},$$

$$\Psi_{\mathrm{rel}} = [\langle\psi(0)\rangle - \langle\psi(L)\rangle]\frac{h}{\sigma N v}.$$

Observables are computed from mean profiles, rather than averaging nonlinear snapshot ratios. Error estimates include recomputed core boundaries, shared bath uncertainty, and potential–height covariance. Γ differs from the actual net charge including C and anions, which is stored separately in the raw data.

Finite-grid and cross-sectional-area controls

Except for the salt-free row, $z_+ = 3$ and the target is $\rho_b = 0.01\,\mathrm{M}$. Chain counts were adjusted with area to retain $\sigma = 0.1\,\mathrm{nm}^{-2}$. Bulk activities were determined independently for each case.

Control	M	A [nm^2]	h [nm]
Standard	400	40	39.28 ± 0.35
Finer grid	800	40	39.18 ± 0.20
Smaller area	400	20	37.85 ± 0.49
Larger area	400	80	39.63 ± 0.19
Salt-free, finer grid	800	40	58.79 ± 0.50

For all three valences, $\overline{\Gamma}_{0.5}$ was positive; potentials at 10 M were within 2 SE of zero. **Limits:** A sets fluctuation strength in this 1D model, so varying area is more than a numerical precision check. A few grid controls do not establish continuum convergence. This restricted model has not reproduced the paper's contraction, re-expansion, and charge/potential reversal from 3D charge correlations. These results also do not validate convergence of the original CL-FTS dynamics. Code: `positive_brush_1d.py`, `run_positive_brush.py`. Numbers, samples, and jobs: `dh_salt_runs/positive_brush_20260907/`. Production FFT/C++/CUDA and the existing CL drift were unchanged in this work. The following pages retain the earlier small-system and SCFT validation records.

Earlier checks: small systems and SCFT

Validation update: 2026-09-07 / `devel/charged_polymers`

Earlier validation verdict
The real-field/particle sampler for a small finite 1D system is consistent with reference values. PRL Figs. 1–3 were recalculated, but quantitative discrepancies remain.

Models and scope of validation

The 1D sampling in this section checks the existing finite 9-bead action. The PRL calculations use a separate SCFT with $N = 100$ chains, point ions, and incompressible P/S. These results do not validate the accuracy of the existing complex Langevin (CL) dynamics or establish full PRL reproduction.

After exactly integrating the repulsive auxiliary field and electrostatic field ψ, a Gibbs sampler retains particle positions and the real field in the attractive channel. Fields and positions are drawn from their conditional distributions; integrating the field recovers the original particle action. The next page gives this relation and the conditional mean and covariance.

Controls for the same four-cell model

Polymer density ϕ_P in cell 1 for $M = 4$, $E = 25$. Values following $\pm$ are standard errors (SE), accounting for blocking and autocorrelation.

Calculation	Mean ± SE
Exact particle sum	0.47375505
Real-field/particle Gibbs, seed 1101	$0.47113827 \pm 0.00095049$
Real-field/particle Gibbs, seed 1201	$0.47464444 \pm 0.00083619$
Particle-only Gibbs, seed 2101	$0.47479012 \pm 0.00115897$
Historical native CL, seed 71	$0.45115654 \pm 0.00333992$
Historical native CL, seed 72	$0.44947711 \pm 0.00337511$

The exact result sums $4^8 = 65{,}536$ particle configurations of the same action. Historical CL values are stored results at $\Delta t = 0.01$; the four-cell CL $\Delta t \to 0$ limit was not newly investigated here.

Completed calculations and the historical report

There are 21 completed 1D sampling runs and 25 converged PRL Fig. 1 results. The Fig. 2–3 salt scans and unit/grid checks are appended below. Records of three initial failed Fig. 1 iterations are preserved; subsequent Newton/continuation runs converged for the same target equations.

The original report appended to the combined PDF is historical. Its claims of "complete quantitative reproduction," "exact" CL, and discrepancies due solely to LDA are superseded by the current validation status. Production FFT and CL drift were unchanged during these checks.

Real-field integration and sampling

Let $\boldsymbol{R}$ denote particle configurations and $\boldsymbol{x}$ a real field on M cells. $\boldsymbol{\ell}(\boldsymbol{R})$ is the attractive-channel combination of smeared densities with its uniform mode removed. Define the spatial mean and uniform-mode projection by

$$\overline{f} = \frac{1}{M}\sum_{i=1}^{M} f_i, \qquad \overline{x} = \overline{\ell} = 0, \qquad \Pi = I_M - \frac{\mathbf{1}\mathbf{1}^{\mathsf{T}}}{M}. \tag{1}$$

Here $\mathbf{1}$ is the all-ones vector and Π projects onto the zero-mean subspace. $a > 0$ is the Gaussian coefficient of the attractive auxiliary field; $C > 0$ is the action prefactor. The a on this page is distinct from the PRL Born radius.

A joint distribution preserving the original particle weight

Writing the original particle energy as $U_{\mathrm{particle}}(\boldsymbol{R})$ and the chain bond weight as $B(\boldsymbol{R})$ gives

$$U_{\mathrm{rep}}(\boldsymbol{R}) = U_{\mathrm{particle}}(\boldsymbol{R}) + \frac{C}{4a}\,\overline{\ell(\boldsymbol{R})^2}, \tag{2}$$

$$P(\boldsymbol{R}, \boldsymbol{x}) \propto B(\boldsymbol{R})\, \exp\left[-U_{\mathrm{rep}}(\boldsymbol{R}) - C\,\overline{ax^2 + \ell(\boldsymbol{R})x}\right]. \tag{3}$$

Completing the square,

$$ax_i^2 + \ell_i x_i = a\left(x_i + \frac{\ell_i}{2a}\right)^2 - \frac{\ell_i^2}{4a}. \tag{4}$$

integrating over $\mathcal{X} = \{\boldsymbol{x} : \mathbf{1}^{\mathsf{T}}\boldsymbol{x} = 0\}$ therefore gives, up to a normalization independent of $\boldsymbol{R}$,

$$\int_{\mathcal{X}} P(\boldsymbol{R}, \boldsymbol{x})\, \mathrm{d}\boldsymbol{x} \propto B(\boldsymbol{R}) \exp[-U_{\mathrm{particle}}(\boldsymbol{R})]. \tag{5}$$

The added quadratic term cancels exactly. This identity represents the same finite action; it does not guarantee exact averages from a finite Monte Carlo trajectory.

Field and particle updates

For a fixed particle configuration, the Gaussian conditional distribution has

$$\mathbb{E}[\boldsymbol{x} \mid \boldsymbol{R}] = -\frac{\boldsymbol{\ell}(\boldsymbol{R})}{2a}, \qquad \mathrm{Cov}(\boldsymbol{x} \mid \boldsymbol{R}) = \frac{M}{2Ca}\,\Pi. \tag{6}$$

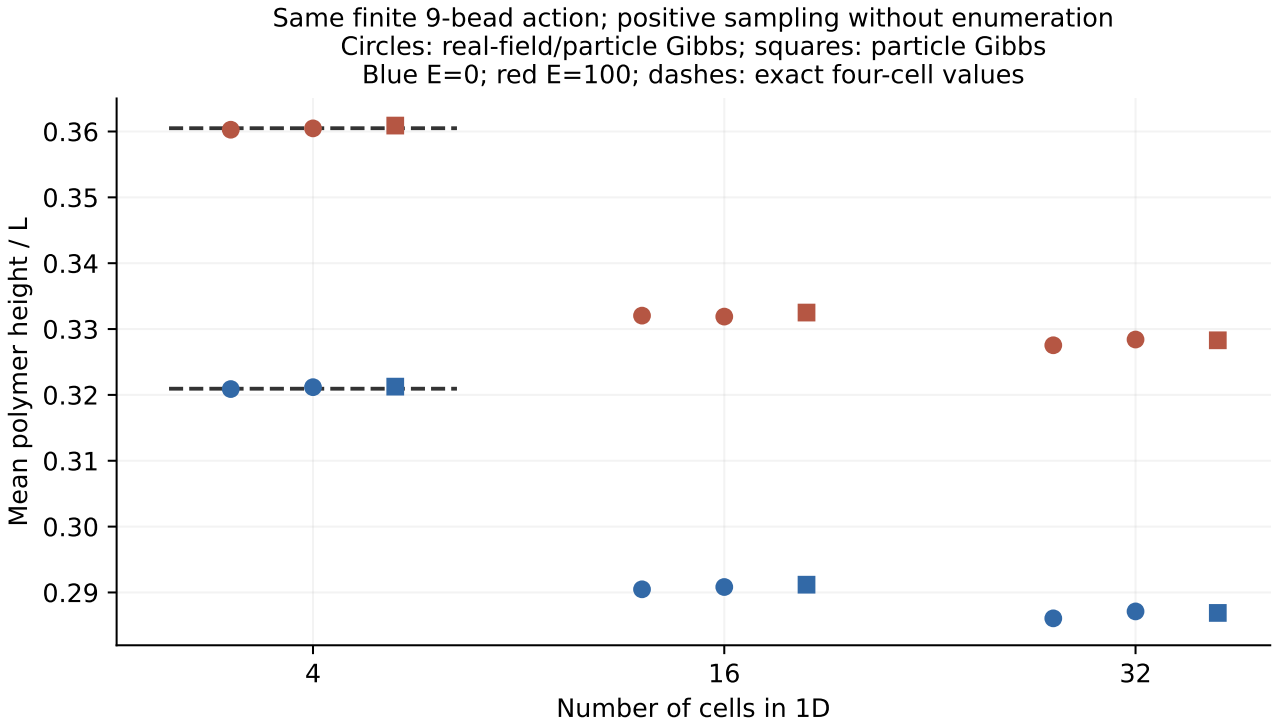
In practice, draw $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, I_M)$ and set $\boldsymbol{x} = -\boldsymbol{\ell}/(2a) + \sqrt{M/(2Ca)}\,\Pi\boldsymbol{\eta}$. At fixed field, the conditional probability for a free particle at candidate position r is

$$P(R_j = r \mid \boldsymbol{R}_{-j}, \boldsymbol{x}) = \frac{B(\boldsymbol{R}^{j\to r}) \exp[-H(\boldsymbol{R}^{j\to r}, \boldsymbol{x})]}{\displaystyle\sum_{s=1}^{M} B(\boldsymbol{R}^{j\to s}) \exp[-H(\boldsymbol{R}^{j\to s}, \boldsymbol{x})]}, \qquad H = U_{\mathrm{rep}} + C\overline{ax^2 + \ell x}. \tag{7}$$

Each update sums only M candidate positions, including all interactions and self terms, neighboring bead bond weights, and the fixed first bead. Sampling requires no enumeration of all M^8 configurations. The particle-only Gibbs control also integrates out the attractive field.

Larger 1D grids and reference checks

Parameters: $N_{\mathrm{ref}} = 4$, 9 beads in total, $L = 1$, $\chi N = 2$, $\zeta N = 4$, and smearing radius 0.1. The first bead is fixed in the first cell; bonds are reflecting. Each run uses 2×10^6 sweeps, discards the first 2×10^5, and stores every 10 sweeps, yielding 1.8×10^5 samples.



Mean height $\langle z \rangle / L$ at $E = 100$

Cells	Field/particle 1101	Field/particle 1201	Particle-only 2101
4	0.360268 ± 0.000306	0.360473 ± 0.000374	0.360899 ± 0.000341
16	0.332038 ± 0.000318	0.331891 ± 0.000292	0.332504 ± 0.000319
32	0.327538 ± 0.000321	0.328409 ± 0.000294	0.328293 ± 0.000331

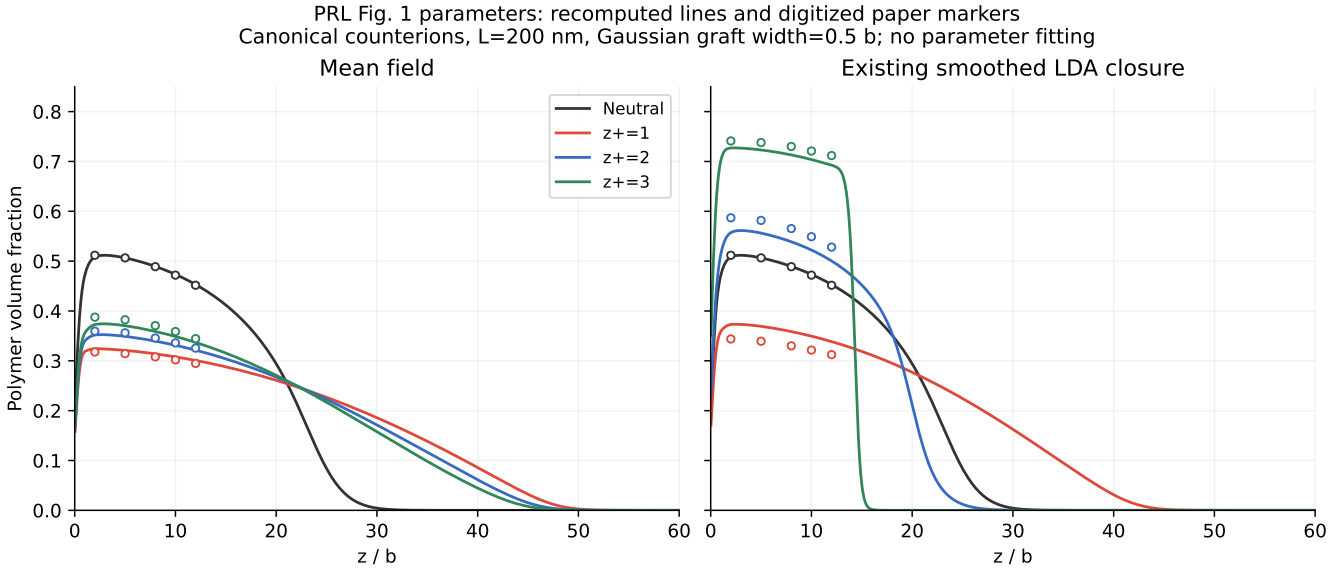
The exact four-cell height is 0.36049837. In all nine $M = 4$ runs, every measured quantity was within 2.781 SE of its exact value. The largest difference between the two samplers across all runs was 3.049 combined SE. These are descriptive statistics over many correlated positions and observables, not a claim that every individual 2 SE check passed.

No exhaustive reference is available at $M = 16, 32$. The two positive-probability samplers were compared. The $16 \to 32$-cell height change exceeds statistical errors, so continuum convergence remains unverified. The spatial representation of the first-cell anchor also changes with the grid.

Independent checks: Gaussian-integration energy identity error $\leq 1.25 \times 10^{-14}$; conditional probability error for fields with standard deviation $5 \leq 7.78 \times 10^{-16}$; species mass conserved in every sample. Local conditional field moments and the sampled first cosine mode's mean and square were also checked. These small 9-bead results do not validate performance for an actual $N = 100$ brush.

PRL Fig. 1: recalculated curves

Paper parameters: $N = 100$, $b = 1\,\mathrm{nm}$, $v = 1\,\mathrm{nm}^3$, $\sigma = 0.1\,\mathrm{nm}^{-2}$, $a = 0.25\,\mathrm{nm}$, $\ell_\mathrm{B} = 0.7\,\mathrm{nm}$, $\chi = 0.5$, and $I_\mathrm{b} = 0.3\,\mathrm{M}$. Primary runs use canonical monovalent counterions, GC salt, and $L = 200\,\mathrm{nm}$. The graft is Gaussian with width $0.5b$; bonds use a truncated bottom and reflecting top.



Polymer volume fraction ϕ_P at $z/b = 5$

Curve	Paper (approx.)	This calculation, $N_z = 3200$
Neutral	0.5065	0.5066
$z_+ = 1$, mean field	0.3144	0.3210
$z_+ = 2$, mean field	0.3562	0.3491
$z_+ = 3$, mean field	0.3824	0.3705
$z_+ = 1$, LDA correlations	0.3392	0.3692
$z_+ = 2$, LDA correlations	0.5817	0.5562
$z_+ = 3$, LDA correlations	0.7379	0.7231

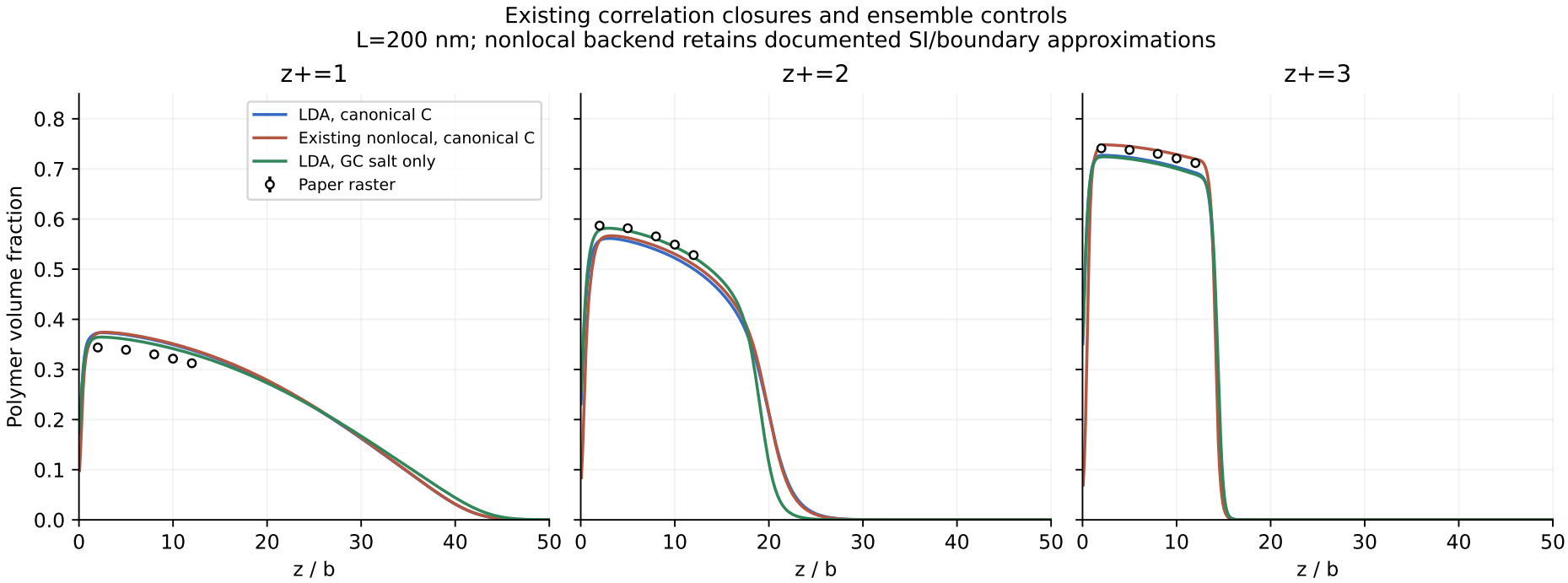
Markers were read from the original PDF raster at $z/b = 2, 5, 8, 10, 12$. Estimated digitization uncertainty is about ± 0.004 in ϕ, **not a statistical error**. These are neither the authors' raw data nor an error assessment of the entire curves.

The maximum differences at the five interior positions for the three LDA curves are 0.0300, 0.0298, and 0.0192. At those positions, changes from $N_z = 1600 \to 3200$ are at most 1.98×10^{-5} in every case. The neutral curve and multivalent collapse trend agree, but the observed grid error does not explain the discrepancies in the correlated curves.

Equal Born radii give $u_K = z_K^2 U$, enabling coupled Newton equations for $V = W/N$ and U. Convergence requires every residual of W, Poisson, self-energy, and volume constraints to be below 10^{-7}. Independent chain density difference $\leq 5.67 \times 10^{-15}$; contour partition error $\leq 4.96 \times 10^{-14}$. The numerical graft/wall representation is not assumed identical to the paper's δ graft.

Effects of correlation closure and ensemble

Controls use the same Fig. 1 parameters. Neither LDA nor the current nonlocal backend has been validated as an implementation of the full SI theory.



Maximum $|\Delta\phi|$ from the paper: five interior positions

z_+	LDA + C	LDA GC-only	nonlocal + C
1	0.0300	0.0212	0.0315
2	0.0298	0.0088	0.0301
3	0.0192	0.0228	0.0081

C denotes canonical monovalent counterions. GC-only mean field agrees to within 0.0027 at the same positions. For $L = 200 \to 400$ nm, the LDA height $2\langle z\rangle$ changes by $30.9787 \to 31.3638$ nm at $z_+ = 1$ and $14.3395 \to 14.3667$ nm at $z_+ = 3$. This moment height is not the Gibbs height of Fig. 2.

Remaining discrepancies and reproduction scope

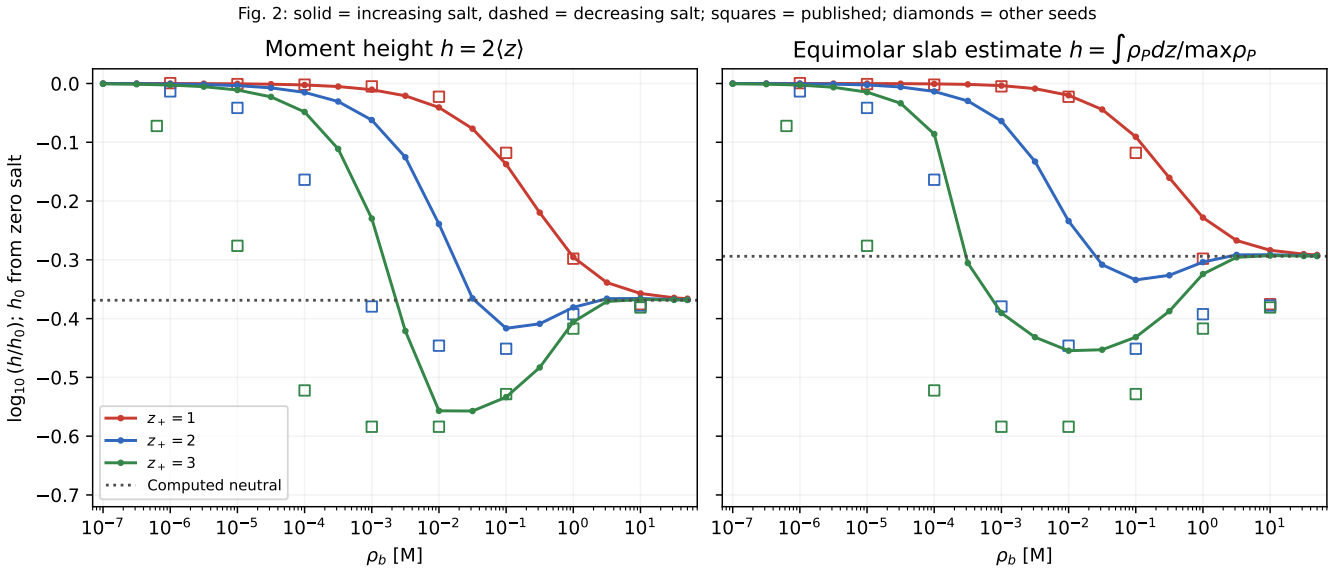
Trivalent nonlocal SCFT converged after continuation, with residuals below 4.61×10^{-9}. Omitted I_{ex}, Green-function boundary and smearing approximations, and quadrature/grid validation remain unresolved. Quantitative reproduction for all three valences is incomplete.

This page checks Fig. 1. New Fig. 2–3 calculations follow; Fig. 4 was not recalculated in this work. Details: `ONED_PRL_REPRODUCTION.md`. Raw data are in `~/polymer/dh_salt_runs/oned_prl_20260907/`: `comparison.json`, `prl_comparison.csv`, run scripts, and SLURM logs. Paper: Duan et al., PRL 134, 048101 (2025).

Fig. 2: contraction and re-expansion

Additional calculations: 1D smoothed-LDA SCFT at fixed parameters. These do not validate the paper's full SI theory or the existing CL dynamics.

For each $z_+ = 1, 2, 3$, 19 standard points span $10^{-7} \leq \rho_b \leq 50\,\mathrm{M}$; validated solutions were obtained at 57/57 points. Upward/downward concentration continuation and intermediate points used separate SLURM jobs. No valence-dependent background salt concentration or height correction was fitted.



Solid/dotted: upward/downward scans. Open squares: digitized paper theory curves. Diamonds mark points obtained only from separate initial states, if any. Experimental symbols were excluded. Digitization uncertainty is about 0.01 in $\log_{10}(h/h_0)$, not a statistical error.

Explicit height definitions and salt-free references

$$h_\mathrm{m} = \frac{2\int z\rho_\mathrm{P}(z)\,dz}{\sigma N}, \qquad h_\mathrm{eq} = \frac{\sigma N}{\max_z \rho_\mathrm{P}(z)}, \qquad \int \rho_\mathrm{P}(z)\,dz = \sigma N = 10\,\mathrm{nm}^{-2}.$$

$h_{0,\mathrm{m}}$ =50.91 nm, $h_{0,\mathrm{eq}}$ =38.45 nm were obtained by solving $\rho_b = 0$ directly. The neutral brush has moment height 21.78 nm. The paper mentions a Gibbs dividing surface without specifying the extraction procedure or reference density. Neither definition above is therefore assumed equal to the paper's Gibbs height.

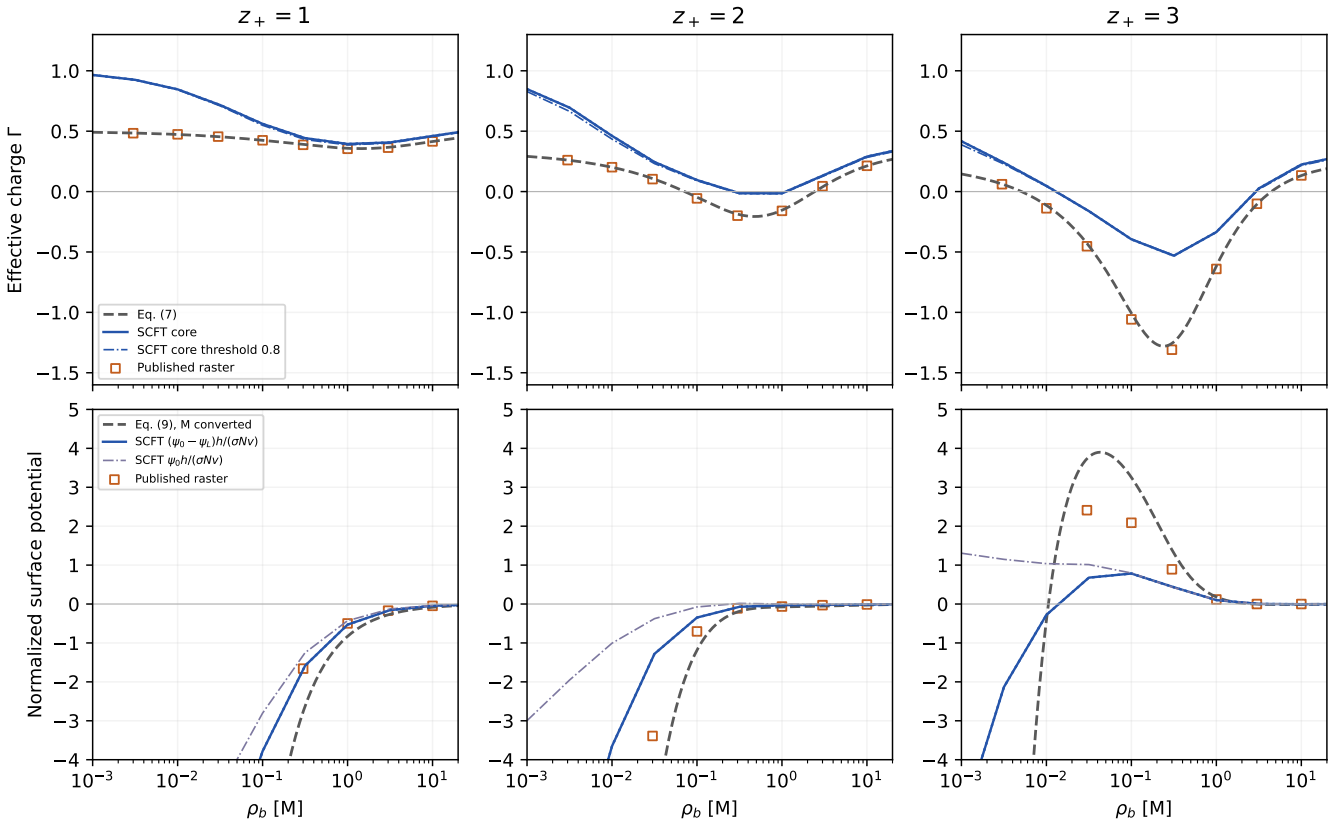
Minimum moment heights among the 19 standard points

z_+	ρ_b [M]	h_m [nm]	$\log_{10}(h_\mathrm{m}/h_{0,\mathrm{m}})$
1	50	21.91	−0.3661
2	0.1	19.51	−0.4165
3	0.03162	14.11	−0.5572

These are sampled minima, not exact minima over continuous concentration. The monovalent minimum is at the scan limit. Multivalent contraction and re-expansion occur, but the onset of contraction at low salt differs from the paper. Changing the height definition does not produce simultaneous quantitative agreement for all three curves.

Fig. 3: effective charge and potential

SCFT profile observables are distinguished from direct evaluation of the paper's Eqs. (7) and (9).



Gray dashed: analytical formulas. Squares: paper curves. Blue: SCFT. Solid/dotted SCFT lines show upward/downward scans. Large negative potentials outside the plotting range are clipped.

Observables measured from SCFT

$$\Gamma(z) = 1 - z_+ \frac{\rho_+(z) - \rho_b}{\rho_P(z)}, \qquad \overline{\Gamma}_c = \frac{\int_{\rho_P > c\rho_P^{\max}} \rho_P(z)\Gamma(z)\,dz}{\int_{\rho_P > c\rho_P^{\max}} \rho_P(z)\,dz},$$

$$\Psi_{\rm rel} = [\psi(0) - \psi(L)]\frac{h_m}{\sigma N v}, \qquad \Psi_0 = \psi(0)\frac{h_m}{\sigma N v}.$$

Both core thresholds, $c = 0.5$ and 0.8, are plotted. These explicit averages of nonuniform profiles are not automatically equivalent to the paper's uniform-brush scaling variables.

The finite box contains a fixed number of monovalent counterions, allowing $\psi(L) \neq 0$. We therefore record both Ψ_0 relative to zero and the wall–far-end difference $\Psi_{\rm rel}$. The sign of $\psi(0)$ alone does not establish charge inversion, particularly at low salt. Wall/far-end potentials are approximated by first/last cell-center values.

Analytical formulas and SCFT have different scopes

SI Eq. (S34) assumes high salt, $\rho_b \gg \rho_P$, with negligible canonical counterion effects. Low-salt SCFT does not satisfy these conditions. Reproducing the analytical formulas therefore does not validate SCFT or CL. The current LDA must be compared with the paper's nonlocal Green function/$I^{\rm ex}$ treatment. The paper also neglects $u_P^{\rm ex}$ for brushes; that omission alone is not an established cause of the discrepancy.

Among the 19 standard points, the divalent core Γ reaches −0.0140, while $\Psi_{\rm rel}$ remains negative. For trivalent salt, $\Gamma_{\min} \simeq -0.5314$ and $\Psi_{\rm rel,max} \simeq +0.7834$. Overcharging and potential reversal are qualitatively distinct, but the trivalent potential peak differs from the paper's value of about 2.5.

Checking Fig. 3 potential normalization

A diagnostic of the equation–figure discrepancy. The scale factor below was not applied to SCFT results.

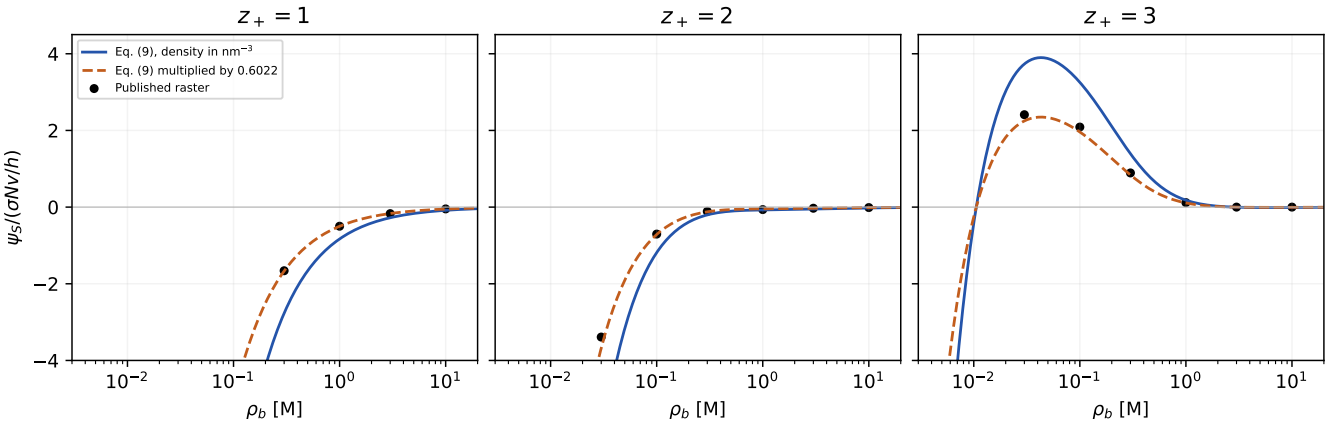
The paper's Gaussian charge-spread function and Eqs. (7), (9) were evaluated as follows.

$$u(x) = 1 - x\ \mathrm{erfcx}(x/\sqrt{\pi}), \qquad u'(x) = -\left(1 + \frac{2x^2}{\pi}\right)\mathrm{erfcx}(x/\sqrt{\pi}) + \frac{2x}{\pi},$$

$$\kappa_b^2 = 4\pi\ell_B z_+(z_+ + 1)\rho_b, \qquad R = -\frac{z_+(z_+ + 1)\kappa_b\ell_B}{4}u'(a\kappa_b),$$

$$\Gamma = \frac{1 - R}{z_+ + 1 - R}, \qquad \Psi_{\mathrm{Eq.9}} = -\frac{(z_+^2 - 1)\Gamma + 1}{z_+^2(z_+ + 1)v\rho_b}.$$

Parameters are $a = 0.25\,\mathrm{nm}$, $\ell_B = 0.7\,\mathrm{nm}$, and $v = 1\,\mathrm{nm}^3$. The formulas use number density $\rho_b[\mathrm{nm}^{-3}] = 0.6022\,\rho_b[\mathrm{M}]$.



Blue: Eq. (9) with unit conversion. Orange dotted: that value multiplied by 0.6022, for diagnosis only. Black dots: digitized original raster. The latter scale equals omitting the density conversion in the denominator; it is not asserted to be the correct physical normalization.

The monovalent check is independent of correlation parameters

For $z_+ = 1$, $\Psi_{\mathrm{Eq.9}} = -1/(2v\rho_b)$, eliminating a, ℓ_B, Γ.

ρ_b [M]	Eq. (9), converted	Paper (approx.)	Eq. (9)×0.6022
0.3	−2.7676	−1.6598	−1.6667
1	−0.8303	−0.4985	−0.5000
3	−0.2768	−0.1701	−0.1667

The monovalent discrepancy is consistent with the M–nm^{-3} conversion factor. Multivalent discrepancies remain. **The units/normalization connecting the paper's equations and figure require confirmation.** The cause is not established without the authors' raw data or plotting code. Source: Duan et al., PRL 134, 048101 (2025), Eqs. (7), (9), Fig. 3, and SI Eqs. (S33), (S38). The printed SI Eq. (S42) lacks the ρ_b denominator visible in main-text Eq. (9); potentials were therefore evaluated from the main-text equation.

Numerical checks and remaining gaps

Numerical issues addressed in a separate Python research runner

The existing real-space chain calculation also suffered floating-point underflow at low salt. Densities differed from an independent log-space calculation by up to $0.2098\,\mathrm{nm}^{-3}$, hidden by per-bead floors and final mass normalization. The new runner evaluates positive bond convolutions in extended precision, using one partition function for every contour position. Independent log-sum-exp calculations check final densities and contour consistency.

The runner adds Poisson–Boltzmann Newton updates including derivatives of canonical C normalization. The Jacobian is tridiagonal plus a rank-one term. For random fields (std $\simeq 5$), its relative finite-difference error was 3.77×10^{-10}. These changes are confined to `run_prl_fig23_checked.py`. Production FFT/C++/CUDA, CL drift, and the existing `prl_scft.py` were unchanged in this work.

Acceptance criteria and maximum observed errors

Check	Required	Observed maximum
Full SCFT residual	10^{-7}	6.02×10^{-8}
Independent chain density difference	10^{-10}	2.33×10^{-13}
Relative contour partition error	10^{-10}	1.98×10^{-12}
Polymer integral error [nm^{-2}]	10^{-10}	3.55×10^{-15}
Canonical C integral error [nm^{-2}]	10^{-10}	3.55×10^{-15}

215 extended-precision runs passed validation. Failed attempts are excluded from successful results; their logs and checkpoints are preserved. Across 57 upward/downward pairs at matching concentrations, the largest height difference is $2.85\times10^{-7}\,\mathrm{nm}$. The maximum full polymer density profile difference is $1.90\times10^{-8}\,\mathrm{nm}^{-3}$. This checks iterative solution consistency; it does not prove a free-energy minimum.

Grid and box size: moment height changes [nm]

The reference is $L = 200\,\mathrm{nm}$, $N_z = 1600$. Grid refinement uses $N_z = 3200$; box enlargement uses $L = 400\,\mathrm{nm}$, $N_z = 3200$.

z_+	ρ_b [M]	Δh, finer grid	Δh, larger box
1	0	8.02×10^{-4}	0.01049
1	1.00×10^{-7}	8.02×10^{-4}	0.009946
2	0.1	0.00221	−0.5041
3	0.01	0.003446	−0.3023
3	10	0.001948	-5.11×10^{-5}

These are selected-point checks, not proof of $L\to\infty$ convergence for the full scan. One failed high-salt PB residual check was rerun with a tighter inner PB tolerance. The SCFT acceptance threshold was not relaxed.

Raw data: `~/polymer/dh_salt_runs/fig23_prl_20260907/`. Preserved files include `comparison.json`, `fig23_primary.csv`, each job's `run.sh`, `summary.json`, `profiles.npz`, SLURM logs, and digitized figure coordinates. Reproduction steps and limitations are recorded in `FIG23_PRL_REPRODUCTION.md`.

Side-by-side comparison: Duan, Agrawal & Wang, PRL **134**, 048101 (2025) vs. this codebase: (i) their SCFT reimplemented, (ii) charged CL-FTS

Internal research note (devel/charged_polymers) — figures from the cited paper are reproduced here for private scholarly comparison only

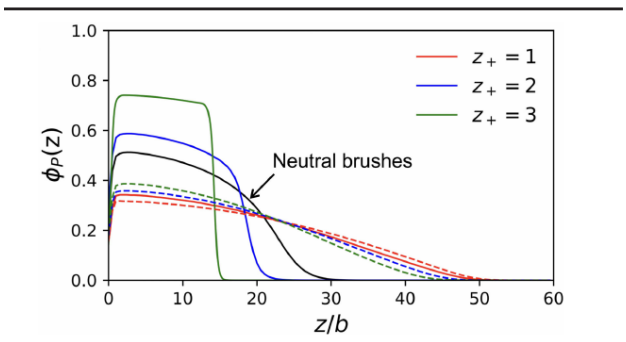
2026-08-26

Abstract

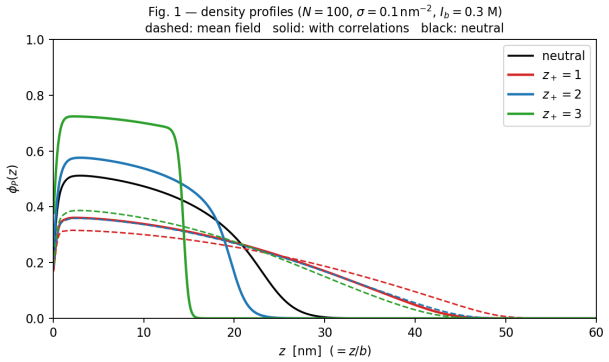
Panel-by-panel comparison of the polyelectrolyte-brush results of Duan, Agrawal & Wang [PRL **134**, 048101 (2025)] — obtained with their electrostatic-correlation-augmented SCFT — against our independent reproduction with fully fluctuating charged complex-Langevin FTS (smeared-charge compressible model, wall boxes, discrete chains). Left panels: the paper's figures. Right panels: our measurements at the strongest coupling inside our CL-stable envelope (z_+=3 ion correlation $u_3 \approx 13\,k_BT$, Θ solvent, $\alpha = 1$ fully charged backbone, $N = 25$, $\bar{N} = 10^4$). Full provenance: `RESULTS_FIG1_REPRO.md` §3–3f; all numbers audited against the run data.

Part I — their own theory, our implementation (correlation SCFT)

Direct reimplementation of their variational/self-energy SCFT (`prl_scft.py`): repo discrete-chain propagator on a real-space bond-convolution backend, guarded-Newton nonlinear PB with analytic grand-canonical ions, LDA self-energies (SI Eq. S33), exact ξ elimination, α- and correlation-strength continuation. All at their exact Fig. 1 parameters.



(a) PRL Fig. 1.

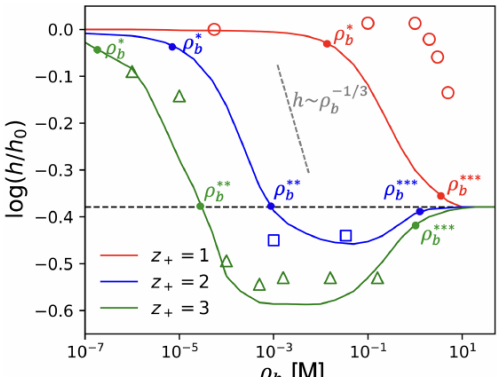


(b) Same theory, our implementation.

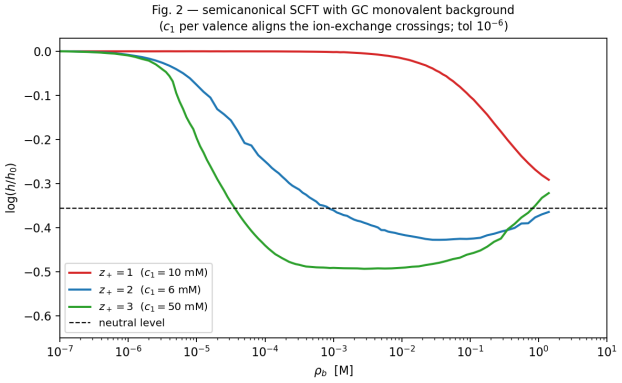
Verdict: reproduced (quantitative). Curve-for-curve agreement, all states converged to 5×10^{-4}, including the near-wall entropic depletion layer (hard-wall truncated bond integral; graft source a Gaussian of width $0.5b$ centered on the wall (half-Gaussian, the smooth analog of their $\delta(z)$; finite in the discrete hard-wall formulation)): mean-field dashed family ($\phi \approx 0.31$–0.40, extent 45–52 nm, nearly valence-blind; theirs 0.32–0.38, 45–50b); neutral ($h = 21.8$ nm, plateau 0.51, edge $\sim 28b$; theirs ~ 0.51, $\sim 28b$); z_+=2 collapsed plateau $\phi = 0.58$ to 18.7 nm (theirs ~ 0.59 to $\sim 19b$) with $\Gamma = -0.056$ (their algebraic Eq. 7 value: -0.048) and $\psi_S < 0$; z_+=3 plateau $\phi = 0.72$–0.73, $h = 14.5$ nm (theirs ~ 0.74 to $\sim 14b$) with $\Gamma = -0.32$ and $\psi_S = +0.66$ (charge inversion); z_+=1 slightly compacted, no overcharging. The Fig. 3 sign structure (z_+=2: overcharging without ψ_S inversion; z_+=3: both)

is reproduced exactly. One physics note: the pointwise LDA self-energy has no wavelength cutoff and goes grid-scale-unstable at z_+=3 full coupling; smoothing the local ionic strength over the Born radius a (the regularization their nonlocal G_s carries intrinsically) restores clean convergence and leaves z_+=1, 2 untouched. Full account: `RESULTS_PRL_SCFT.md`.

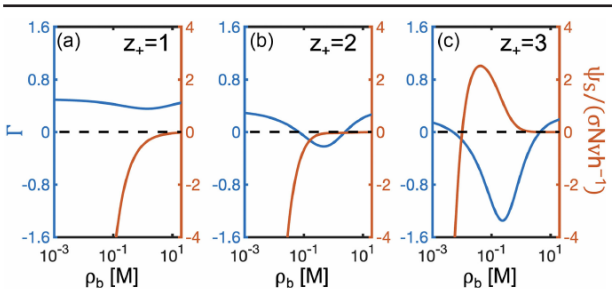
Their Figs. 2–4 with the reimplemented SCFT



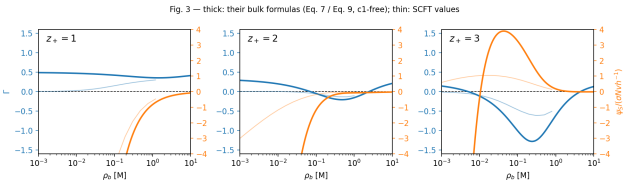
(a) PRL Fig. 2.



(b) Ours, semicanonical with a GC monovalent background c_1 (the physical form of the counterion reservoir; the onset position is reservoir-controlled — shown by the $1/L$ box scaling — and the paper reports neither box nor background). c_1 per valence aligns the ion-exchange crossings (z_+=2: 9.2×10^{-4} vs their $\sim 10^{-3}$ M; z_+=3: 2.1×10^{-5} vs their $\sim 3 \times 10^{-5}$ M); valleys $-0.42/-0.49$ (theirs $-0.45/-0.58$; the z_+=3 background screening shallows it). A single c_1 cannot align both (their crossing ratio 33 vs this tier's ∼14 — the I_{ex} signature).



(a) PRL Fig. 3.



(b) Ours. Thick blue: their Eq. 7 (z_+=3 min -1.27 vs their -1.3); thick orange: their Eq. 9 (their orange curve is this Donnan estimate, not a wall potential — verified by shape). The z_+=3 peak: Eq. 9 with our Γ gives +3.9, the true SCFT wall value +0.9; their +2.5 lies between and is NOT reproducible from their own published Γ + Eq. 9 (which would give +5) — an internal inconsistency traceable only to their unpublished numerics.

Part II — fully fluctuating charged CL-FTS vs the paper

Final (2026-08-29): three-way judgment at matched parameters, hard-wall boxes on both sides. Their correlation SCFT (Part I implementation) applied to the CL campaign's own system (Θ, N=25, matched

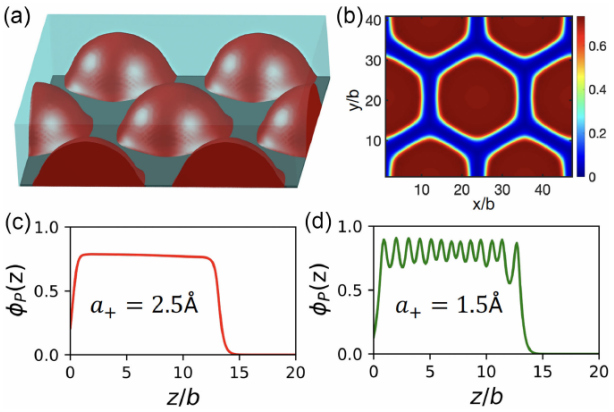
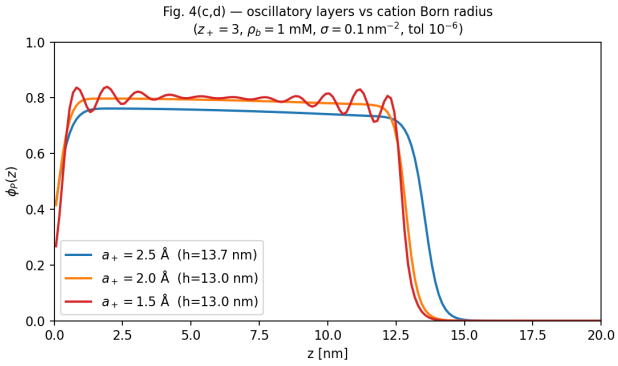
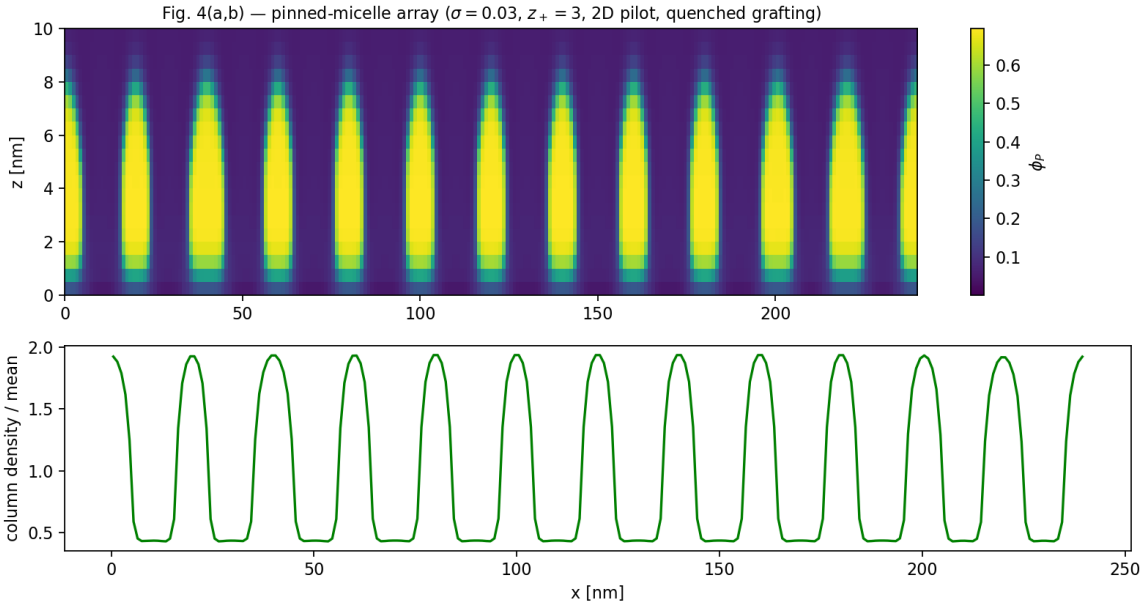


FIG. 4. Microphase separation in the morphology of PE

(a) PRL Fig. 4.



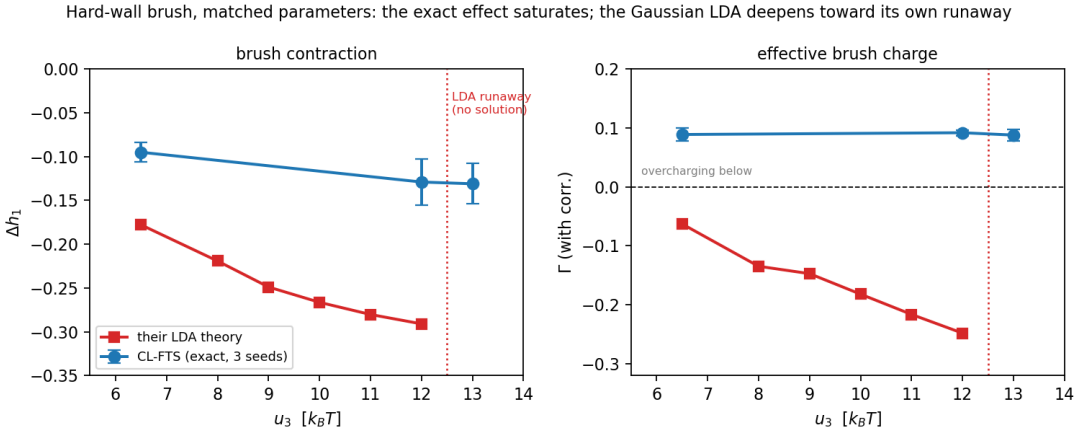
(b) Ours, their (c,d): $a_+ = 2.5, 2.0$ Å give smooth collapsed films; $a_+ = 1.5$ Å gives converged stationary oscillatory layers (period $\sim b$) — the same physics as their panels, drawn as 1D profiles rather than density renders. Their (a,b): **the visual difference is dimensionality** — their hexagonal pinned-micelle lattice is a 3D result, while our calculation (`prl_scft2d.py`) is a 2D (x, z) pilot in which micelles can only appear as stripe cross-sections by construction. The 2D pilot reproduces the lateral microphase separation itself: the uniform film is metastable and seeded runs converge to arrays of pinned dense domes ($\phi \sim 0.75$, spacing 15–30 nm, near-bare gaps; next figure). A full 3D extension would be required for the hexagonal arrangement.



Converged laterally structured state of the quenched 2D pilot at their Fig. 4(a,b) conditions ($\sigma = 0.03\,\mathrm{nm}^{-2}$, z_+=3, $\rho_b = 1$ mM): an array of pinned dense domes ($\phi \sim 0.75$, height ~8 nm, spacing 20 nm, near-bare gaps), coexisting with the metastable uniform film (first-order lateral microphase separation).

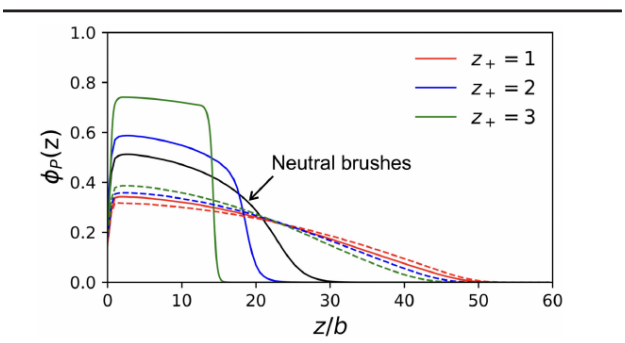
Born radii, matched graft): mean fields agree to 1–6%; the correlation DIFFERENTIALS show the exact effect SATURATING ($\Delta h_1 = -0.095 \to -0.129 \to -0.131$ at $u_3 = 6.5/12/13\,k_BT$) while the Gaussian LDA deepens

toward its own fold ($-0.190 \to -0.291 \to$ no solution): overshoot 2.0× at $6.5\,k_BT$, 2.3× at $12\,k_BT$, and beyond the runaway threshold $u_3^* \approx 12.5\,k_BT$ the LDA electrolyte sector has no physical solution (saturated-condensation branch) while CL remains well-behaved. The LDA's overcharging (Γ to -0.25) is contradicted by three-seed-robust CL positives ($\Gamma \approx +0.09$) at every tested coupling.

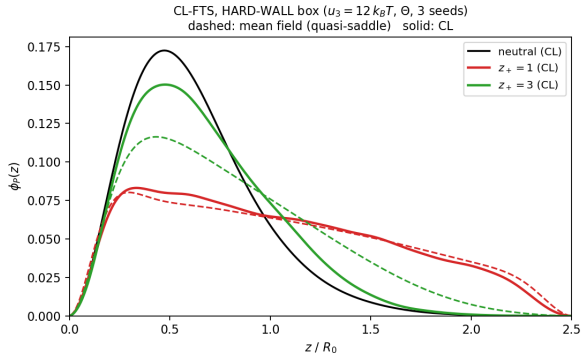


LDA (red, deterministic scan $u_3 = 6.5$–$12\,k_BT$) vs exact CL-FTS (blue, 3 seeds), same hard-wall microscopic system. Dotted line: LDA runaway threshold ($u_3^* \approx 12.5\,k_BT$), beyond which their theory has no physical solution.

Fig. 1 — density profiles: valence-selective correlation collapse



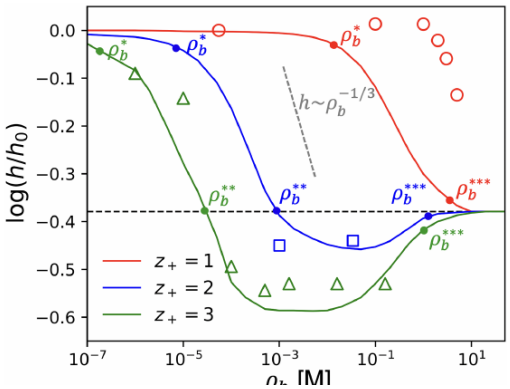
(a) PRL Fig. 1: with correlations (solid), the z_+=3 brush collapses *below the neutral profile*; mean field (dashed) is swollen and nearly z_+-insensitive. (N=100, b=1 nm, a=2.5 Å, I_b=0.3 M.)



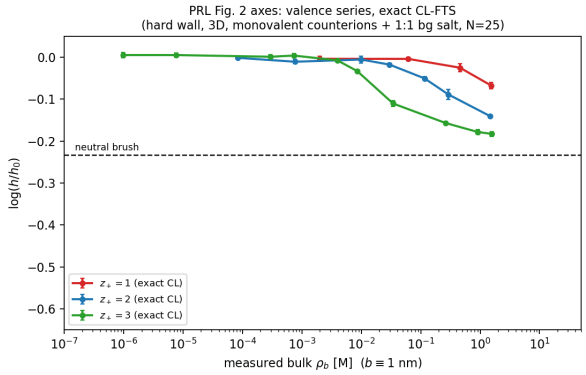
(b) Ours, HARD-WALL rerun (Θ, u_3=12 k_BT, 3 seeds; dashed: quasi-saddle mean field, solid: CL): profiles now show the correct wall depletion layer ($\phi(0) \approx 2\times10^{-4}$). Correlations collapse the z_+=3 brush (h_1: 0.80 → 0.67) while z_+=1 barely moves; the collapsed brush remains *above* our neutral reference (h_1=0.62).

Verdict: reproduced (mechanism). The defining structure — correlation-induced, z_+-selective collapse invisible at mean field — appears identically. The causality was pinned three ways (§3, reflecting-wall era; mechanism unchanged by the wall fix): the contraction scales as z_+^2 (ratio 8.7 ≈ 9), vanishes in the neutral control, and tracks the Coulomb coupling E. **Difference, resolved in §3h:** the paper's collapse undershoots the neutral height; ours saturates well above it. The matched-parameter comparison shows this is not a limitation of our envelope — the exact effect saturates while their Gaussian LDA keeps deepening — so the below-neutral amplitude is an artifact of their approximation.

Fig. 2 — nonmonotonic $h(\rho_b)$: collapse and reexpansion



(a) PRL Fig. 2: $\log(h/h_0)$ vs ρ_b. For z_+=2, 3 the height is nonmonotonic — collapse below the neutral line, then re-expansion — matching the Tirrell-group experiments (symbols).



(b) Ours, on THEIR axis and axes (monovalent counterions + 1:1 background salt, multivalent SALT scanned; hard wall, 3D; x = the MEASURED reservoir concentration of the canonical box, $b \equiv 1\,\mathrm{nm}$): exact CL valence series — common extended plateau, valence-ordered collapse z_+=3 > 2 > 1 (z_+=1 tracks mean field), floor above the neutral brush (dashed), nonmonotonic Γ (§3i-d,g).

Verdict: reproduced (mechanism, on the correct axis) — amplitude gap attributed to the LDA (§3i-b–3i-d). An important correction: our earlier salt scans used trivalent brush COUNTERIONS, an axis on which the brush starts collapsed and no valley can exist by construction. On the paper's actual axis (monovalent counterions, trivalent ions arriving only via salt) the exact ensemble DOES reproduce the Fig. 2 phenomenology: extended → collapse (h_1: 1.07 → 0.70, CL below MF throughout, gap growing with salt) → valley floor, with nonmonotonic Γ. What does NOT appear exactly: the amplitude. Their collapse undershoots the neutral brush by ∼40%; our exact valley floor stays above ours. Attribution (§3i-g, magenta curve above): their Gaussian LDA, run on our system at the matched measured ionic environment, gives the full deep valley — onset one decade earlier, 1.75× deeper, below neutral, with an overcharging window — none of which survives exact sampling. (On the trivalent-counterion axis, figure below, the theory additionally loses its solution entirely at low screening.)

Fig. 3 — effective brush charge Γ: neutralization vs overcharging

Verdict: partial, revised by the hard-wall reruns (§3h–3i). Correlation-driven condensation far beyond mean field is real and now hard-wall-confirmed: Γ drops from the mean-field 0.24–0.30 to a 0^+ plateau (+0.008–0.012) in the low-salt window (§3i), and at fixed s=0.02 the exact ensemble SATURATES at $\Gamma \approx +0.09$ across $u_3 = 6.5$–$13\,k_BT$ (§3h). It never crosses zero. The paper's overcharging window ($\Gamma < 0$) is where their Gaussian LDA is already overshooting ours by 2.0–2.3× and heading into its runaway ($u_3^* \approx 12.5\,k_BT$); we now attribute their $\Gamma < 0$ amplitude to the Gaussian/LDA approximation rather than to physics our envelope "cannot yet reach."

Fig. 4 — lateral microphase separation (precursor only)

Verdict: precursor observed, geometry-limited. The lateral-inhomogeneity indicator rises to ∼10× the neutral baseline for charged brushes, strongest at low salt (their Fig. 4 lives at $\rho_b \sim 1\,\mathrm{mM}$), but a quantitative Fig. 4 comparison requires the wide-box campaign.

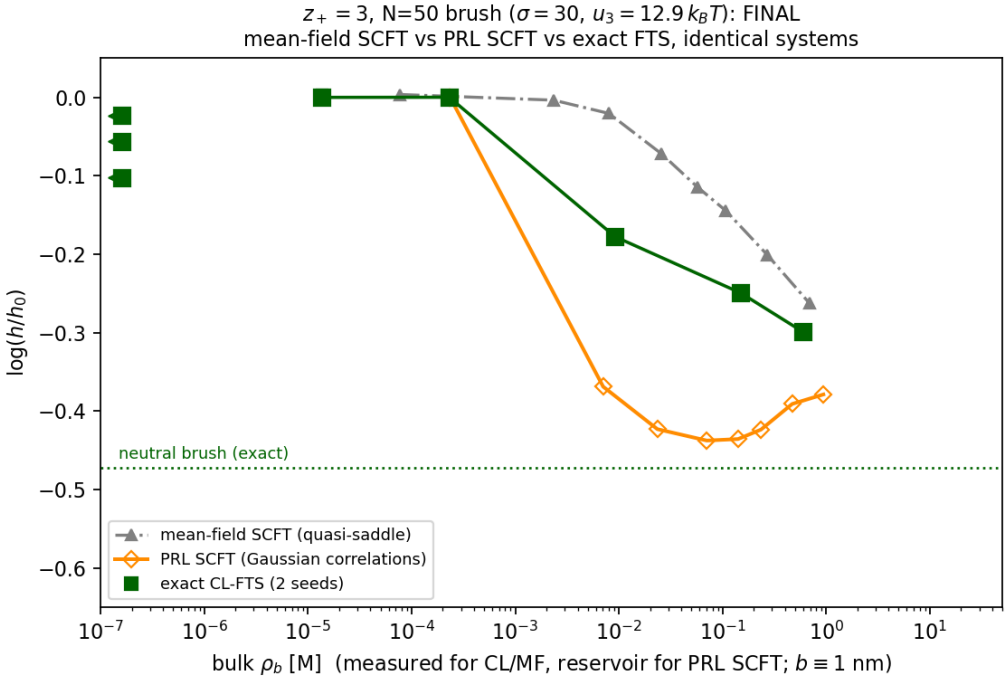


Fig. 2-1 (FINAL) — the z_+=3 three-way judgment at identical parameters (N=50 brush, σ=30, tall box, $u_3 = 12.9\,k_BT$; 8 salt points, 2 seeds). Mean field (gray): monotone. Their PRL SCFT (orange): dives to -0.44 at $\sim 10^{-2}$ M — essentially its neutral line — with overcharging and reexpansion. Exact FTS (green): smooth collapse to -0.30, sitting BETWEEN mean field and their theory at every measurable point (the exact correlation deepening is 40–50% of their prediction), staying 0.17 above the exact neutral brush (dotted, -0.47) with $\Gamma > 0$ throughout ($+0.67 \to +0.11$). Left-arrows: onset-region points whose bulk trivalent is fully depleted by the brush (canonical box; reservoir concentration below detection). The N=25 version is in §3i-g.

	PRL 134, 048101	this work
theory	variational Gaussian (self-energy) SCFT	full CL-FTS sampling
fluctuations	one-loop, self-consistent	all orders (stochastic)
chain	discrete, $N = 100$, $\alpha = 1$	discrete, $N = 25$, $\alpha = 1$
solvent	Θ ($\chi = 0.5$)	Θ ($\chi N = 12.5$)
ions	explicit, Born radius 2.5 Å, grand-canonical	explicit 1-segment, smeared ($a_{\mathrm{ion}} = 0.025$–$0.0125\,R_0$), canonical
z_+=3 ion correlation	$\sim 7\,k_BT$ (near point-like)	$\leq 13\,k_BT$ nominal, cutoff-limited
geometry	1D–3D wall, analytic-ion Poisson	3D all-reflecting box, fluctuating ψ (DCT/ETD)

Correspondence of models and scales

Summary scoreboard

The amplitude gap traces to a single, well-characterized cause: our UV-regularized (smeared-charge) model caps the ion-correlation strength that their near-point-charge treatment attains, and shrinking the smearing collides with a CL stability boundary. Candidate paths beyond it — their grand-canonical analytic-ion formulation (SI Eq. S24) grafted onto our fluctuating-ψ sector, or a dedicated strong-coupling ψ stabilization — are recorded as future work.

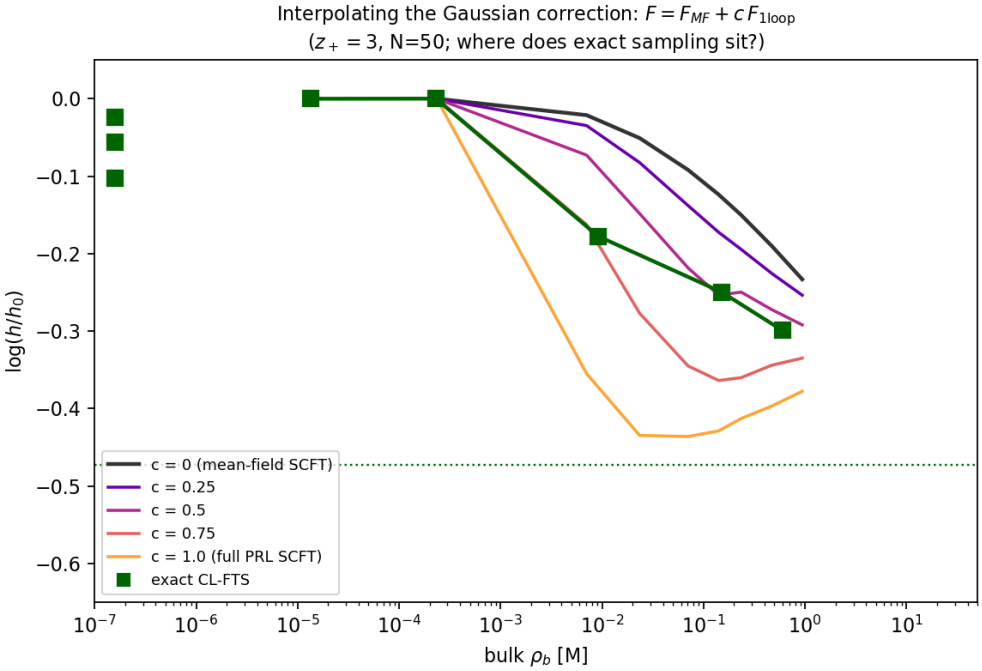


Fig. 2-2 — interpolating their correction, $F = F_{\mathrm{MF}} + c\,F_{1\mathrm{loop}}$: the exact ensemble (green) lies on the c=0.5 member of the family across the whole collapse region. At $u_3 = 12.9\,k_BT$ the non-Gaussian terms cancel HALF of the Gaussian correlation free energy: no physical parameter maps their theory onto exact sampling — only the truncation weight $c \approx 0.5$ does (§3i-i).

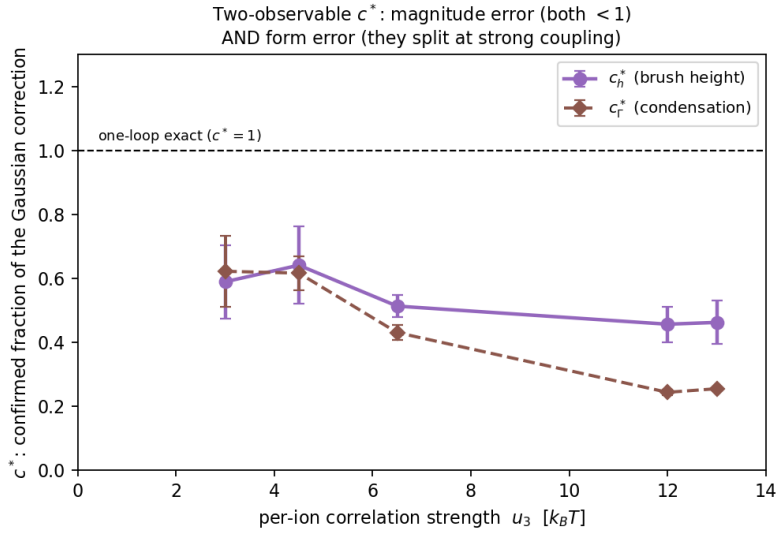
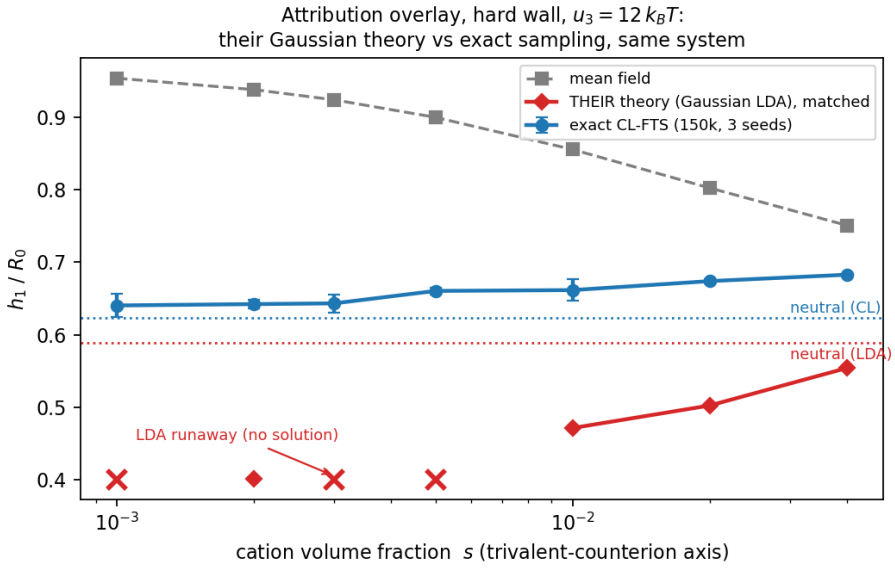
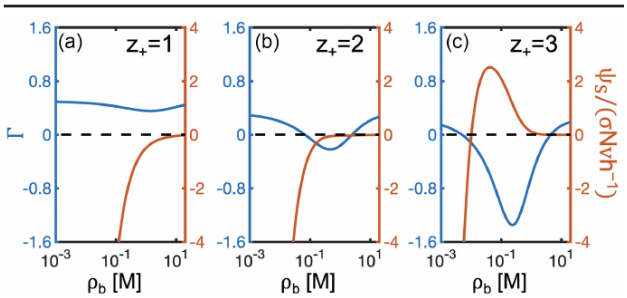


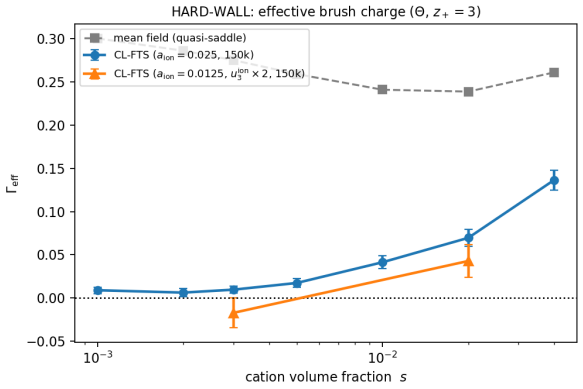
Fig. 2-3 — two-observable $c^*(u_3)$: the fraction of their Gaussian correction confirmed by exact sampling, extracted independently from brush height (c_h^*, purple) and condensation (c_Γ^*, brown) at five coupling rungs (fixed s=0.02). Nowhere is the full correction (c^*=1) supported. At weak coupling both observables give the same $c^* \approx 0.6$ (a genuine effective theory); beyond $u_3 \sim 6\,k_BT$ they SPLIT (0.46 vs 0.24 at u_3=12–13): no single rescaling fixes both observables — the Gaussian truncation carries a form error, consistent with non-Gaussian ion-bridging attraction absent from any single-ion self-energy (§3i-j).



Attribution overlay (trivalent-counterion axis, hard wall, u_3=12 k_BT): their LDA (red) dives below its neutral line and loses its solution (runaway ×); exact CL (blue) saturates at its neutral floor.



(a) PRL Fig. 3: for z_+=2, 3, Γ dips *below zero* in a salt window (overcharging), before recovering at high salt.



(b) Ours, HARD-WALL rerun: correlations drive Γ from the mean-field 0.24–0.30 (dashed) to a 0^+ plateau (complete neutralization, +0.006–0.010 at 150k steps) across the low-salt window, rising to +0.14 at high salt. No blue point crosses zero. Halving the ion smearing (orange, hard-wall, 150k, 2 seeds) pushes the low-salt point to −0.017±0.017 — consistent with complete neutralization, at most marginal overcharging at doubled per-ion coupling.

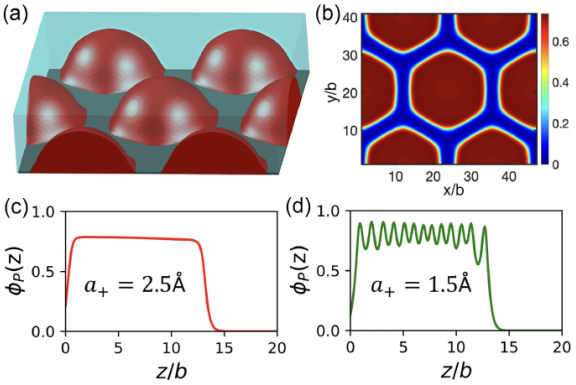
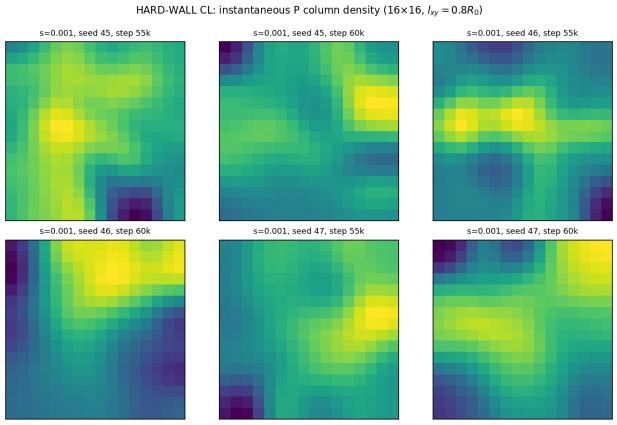


FIG. 4. Microphase separation in the morphology of PE

(a) PRL Fig. 4: pinned micelles in hexagonal arrangement at moderate grafting density; oscillatory layers at high density.



(b) Ours, HARD-WALL rerun (s=0.001, three seeds, two times each): instantaneous P column-density snapshots show box-scale lateral clustering (relative std 0.36–0.44, single cluster per box) — the precursor survives the wall fix — but our lateral box ($0.8\,R_0$) is smaller than the expected micelle spacing, so no lattice can form. A 3–4 R_0 lateral box is the designed follow-up.

PRL result	status	evidence
Part I: their SCFT, our implementation		
Fig. 1 all profiles (MF, corr, neutral)	**reproduced** (quantitative)	curve-for-curve match
Fig. 1–2 collapse below neutral	**reproduced**	z_+=2: 18.7, z_+=3: 14.5 vs neutral 21.8 nm
Fig. 3 overcharging $\Gamma < 0$ (z_+=2, 3)	**reproduced**	-0.057 / -0.32; Eq. 7: -0.048 / -0.65
Fig. 3 ψ_S inversion only at z_+=3	**reproduced**	ψ_S: -0.81 / -0.15 / $+0.75$
Part II: fully fluctuating CL-FTS (honest restatement, 2026-08-29;		
§3–3f were REFLECTING-wall runs, superseded by the hard-wall reruns of §3h–3i)		
Fig. 1 contraction mechanism	**reproduced** (mechanism only)	z_+^2 ratio, E-scaling, neutral null; small amplitude (§3, 3c,
Fig. 2 collapse mechanism (correct axis)	**reproduced**	valence series z_+=1/2/3: common plateau, ordered collap
Fig. 2 below-neutral valley depth	**not reached** (attributed)	matched ionic env.: their theory collapses 1 decade earlie
Fig. 3 strong condensation	**reproduced**	hard-wall: Γ: MF 0.24–0.30 $\to 0^+$ plateau at low salt; sat
Fig. 3 overcharging ($\Gamma < 0$)	**refuted**	exact ensemble stays positive at all tested couplings; LDA
Fig. 1–2 collapse below neutral	**refuted at matched params**	exact contraction saturates far above the paper's amplitu
Fig. 4 lateral micelles	**partial**	box-scale precursor survives hard wall (§3i); dome arrays
Their theory vs exact sampling	—	LDA overshoots 2.0–2.3×, runaway at $u_3^* \approx 12.5\,k_BT$ (§3